



ONE SEMESTER PROJECT

TEST PLAN DOCUMENT

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PROJECT TITLE : SMART BANKLYTICS

An Intelligent Data – Driven Bank Management and Customer Analytics System

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1. INTRODUCTION

1.1 Goals and Objectives

The goal of this test plan is to ensure that the Smart Banklytics system functions correctly, securely, and efficiently according to the Software Requirements Specification (SRS). The objectives include:

- Verifying all functional requirements are correctly implemented
- Ensuring proper handling of valid and invalid inputs
- Validating fraud detection mechanisms
- Confirming role-based access control
- Ensuring accurate analytics and reporting

1.2 Statement of Scope

In Scope

- User Authentication (Login, Logout, Session Management)
- Account Management (Deposit, Withdrawal, Transfer)
- Administrator Functions
- Fraud Detection and Alerts
- Password Recovery
- PDF Statement Generation
- Analytics Dashboard

Out of Scope

- Mobile application testing
- Advanced load/stress testing
- External payment gateway integration

1.3 Major Constraints

- Limited testing time
 - Manual testing approach (no automation tools)
 - Use of SQLite limits large-scale testing
 - No integration with real banking systems
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2.0 TEST PLAN

2.1 Software to be Tested

The software under test is the Smart Banklytics web application. All core modules listed in the scope will be tested. External integrations are excluded.

2.2 Testing Strategy

2.2.1 Unit Testing

Each module (authentication, transactions, fraud detection, etc.) will be tested independently to ensure correctness of individual components.

2.2.2 Integration Testing

Modules such as transaction processing and fraud detection will be tested together to ensure proper interaction and data flow.

2.2.3 Validation Testing

The system will be validated against the SRS to ensure all requirements are met.

2.2.4 High-Order Testing

- Security Testing: Verify unauthorized access is prevented
- Performance Testing: Ensure acceptable response times
- Stress Testing: Evaluate behavior under high input conditions
- Recovery Testing: Ensure system recovery after failure
- User Acceptance Testing (UAT): Validate user satisfaction

2.3 Testing Resources and Staffing

- Tester: System Developer (Student)
- Supervisor: Project Supervisor
- Tools: Browser, local server, database tools

2.4 Test Work Products

- Test Plan Document
- Test Cases
- Test Reports
- Defect Logs

2.5 Test Record Keeping

Test results will be recorded in logs and documented in reports. Screenshots may be used as evidence.

2.6 Test Metrics

- Number of test cases executed
- Number of test cases passed/failed
- Defect count
- Test coverage percentage

2.7 Testing Tools and Environment

- Operating System: Windows/Linux/macOS
- Backend: Python (Flask)
- Database: SQLite
- Frontend: HTML, CSS, JavaScript
- Browser: Chrome, Firefox
- PDF Library: fpdf

2.8 Test Schedule

- Week 1: Unit Testing

- Week 2: Integration Testing
 - Week 3: Validation and System Testing
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3.0 TEST PROCEDURE

3.1 Software to be Tested

Smart Banklytics web application including all functional modules.

3.2 Testing Procedure

Testing will be conducted by executing predefined test cases for each module. Inputs will be provided and outputs compared against expected results.

3.2.1 Unit Test Cases

Each module will be tested independently using defined test cases.

3.2.2 Integration Testing

Modules will be tested together to ensure proper interaction.

3.2.3 Validation Testing

System outputs will be compared with expected results from SRS.

Pass/Fail Criteria

- Pass: Actual result matches expected result
- Fail: Actual result differs from expected result

3.2.4 High-Order Testing

- Security Testing
 - Performance Testing
 - Stress Testing
 - Recovery Testing
 - User Acceptance Testing
-

3.3 Testing Resources and Staffing

- Tester: Student Developer
- Supervisor: Academic Supervisor

3.4 Test Work Products

- Completed test cases
- Test execution reports
- Bug reports

3.5 Test Record Keeping and Test Log

All tests will be logged chronologically with results, including pass/fail status and observations.

4.0 TEST CASES

MODULE 1: USER AUTHENTICATION

Test ID	Test Case	Input	Expected Result	Status
TC-01	Valid login – Customer	Correct username & password	Redirected to customer dashboard	Pass
TC-02	Valid login – Admin	Admin username & password	Redirected to admin dashboard	Pass
TC-03	Invalid login	Wrong username or password	Error message displayed	Pass
TC-04	Empty login fields	Blank username and password	Error message displayed	Pass
TC-05	Session persistence	Navigate after login	User stays logged in	Pass
TC-06	Access dashboard without login	Direct URL access	Redirected to login page	Pass

MODULE 2: CUSTOMER ACCOUNT MANAGEMENT

Test ID	Test Case	Input	Expected Result	Status
TC-07	View account balance	Login as customer	Balance displayed on dashboard	Pass
TC-08	View transaction history	Login as customer	Transaction list shown	Pass
TC-09	View fraud alerts	Login as customer	Alerts listed on dashboard	Pass

MODULE 3: DEPOSIT

Test ID	Test Case	Input	Expected Result	Status
TC-10	Valid deposit	Account number + valid amount	Balance updated, transaction logged	Pass
TC-11	Deposit – zero amount	Amount = 0	Error: amount must be greater than zero	Pass
TC-12	Deposit – negative amount	Amount = -500	Error: invalid amount	Pass
TC-13	Deposit – non-existent account	Invalid account number	Error: account not found	Pass
TC-14	Deposit – fraud threshold	Amount = 100,001	Account suspended, fraud alert created	Pass

MODULE 4: WITHDRAWAL

Test ID	Test Case	Input	Expected Result	Status
TC-15	Valid withdrawal	Sufficient balance + valid amount	Balance reduced, transaction logged	Pass

Test ID	Test Case	Input	Expected Result	Status
TC-16	Withdrawal – insufficient balance	Amount > current balance	Error: insufficient balance	Pass
TC-17	Withdrawal – zero amount	Amount = 0	Error: amount must be greater than zero	Pass
TC-18	Withdrawal – suspended account	Suspended user attempts withdrawal	Error: account suspended	Pass
TC-19	Withdrawal – fraud threshold	Amount = 100,001	Account suspended, fraud alert created	Pass

MODULE 5: TRANSFER

Test ID	Test Case	Input	Expected Result	Status
TC-20	Valid transfer	Valid sender, receiver, and amount	Sender debited, receiver credited, both transactions logged	Pass
TC-21	Transfer – same account	From and to account are identical	Error: sender and receiver must be different	Pass
TC-22	Transfer – invalid recipient	Non-existent recipient account	Error: invalid recipient account	Pass
TC-23	Transfer – insufficient funds	Amount > sender balance	Error: insufficient funds	Pass
TC-24	Transfer – suspended account	Suspended user attempts transfer	Error: account suspended	Pass
TC-25	Transfer – fraud threshold	Amount = 100,001	Account suspended, fraud alert created	Pass

MODULE 6: FRAUD DETECTION

Test ID	Test Case	Input	Expected Result	Status
TC-26	Large transaction rule	Single transaction > KES 100,000	Fraud alert created, account suspended	Pass
TC-27	Rapid transaction rule	3+ transactions within 5 minutes	Fraud alert created, account suspended	Pass
TC-28	Normal transaction	Amount ≤ 100,000, < 3 in 5 min	Transaction processed normally	Pass

MODULE 7: FRAUD ALERT MANAGEMENT (ADMIN)

Test ID	Test Case	Input	Expected Result	Status
TC-29	Mark alert as Legitimate	Admin selects "legitimate"	Transaction processed, account activated	Pass
TC-30	Mark alert as Investigate	Admin selects "investigate"	Alert flagged for investigation, account remains suspended	Pass
TC-31	Mark alert as Freeze	Admin selects "freeze"	Account frozen, alert marked frozen	Pass
TC-32	Customer appeal alert	Customer submits appeal	Alert action updated to "appealed", admin notified	Pass

MODULE 8: ADMINISTRATOR MANAGEMENT

Test ID	Test Case	Input	Expected Result	Status
TC-33	Create new user account	Valid user details submitted	Account created, account number auto-generated	Pass
TC-34	Create user – duplicate username	Existing username used	Error: username already exists	Pass
TC-35	View all users	Admin accesses user list	All registered users displayed	Pass
TC-36	Suspend user account	Admin clicks suspend	Account status changed to suspended	Pass
TC-37	Activate user account	Admin clicks activate	Account status changed to active	Pass
TC-38	Admin deposit to user	Admin deposits to a user account	User balance updated, transaction logged	Pass

MODULE 9: PASSWORD RECOVERY

Test ID	Test Case	Input	Expected Result	Status
TC-39	Valid password reset	Correct username, account number, ID	Password updated successfully	Pass
TC-40	Mismatched passwords	New password ≠ confirm password	Error: passwords do not match	Pass
TC-41	Invalid credentials	Wrong username/account/ID combination	Error: details do not match any account	Pass

MODULE 10: PDF STATEMENT DOWNLOAD

Test ID	Test Case	Input	Expected Result	Status
TC-42	Download all transactions	No filter applied	PDF generated with all transactions	Pass
TC-43	Download – past 30 days filter	Filter = past 30 days	PDF contains only last 30 days of transactions	Pass
TC-44	Download – past 14 days filter	Filter = past 14 days	PDF contains only last 14 days of transactions	Pass
TC-45	Download – custom date range	Custom start and end date	PDF filtered by specified dates	Pass
TC-46	Download – by transaction type	Filter = deposit / withdraw / transfer	PDF shows only the selected type	Pass

MODULE 11: ANALYTICS DASHBOARD (ADMIN)

Test ID	Test Case	Input	Expected Result	Status
TC-47	Total users KPI	Admin views dashboard	Correct total user count displayed	Pass
TC-48	Total transactions KPI	Admin views dashboard	Correct transaction count displayed	Pass
TC-49	Total deposits KPI	Admin views dashboard	Correct sum of all deposits displayed	Pass
TC-50	Total withdrawals KPI	Admin views dashboard	Correct sum of all withdrawals displayed	Pass
TC-51	Total bank balance KPI	Admin views dashboard	Correct sum of all user balances displayed	Pass

Test ID	Test Case	Input	Expected Result	Status
TC-52	Transaction trend chart	Admin views dashboard	Chart renders with correct date-based data	Pass
TC-53	Account type distribution	Admin views dashboard	Pie/bar chart shows correct account type breakdown	Pass

5.0 TEST SUMMARY

Module	Total Test Cases Expected Pass	
Authentication	6	6
Account Management	3	3
Deposit	5	5
Withdrawal	5	5
Transfer	6	6
Fraud Detection	3	3
Fraud Alert Management	4	4
Admin Management	6	6
Password Recovery	3	3
PDF Download	5	5
Analytics Dashboard	7	7
TOTAL	53	53

A total of 53 test cases were designed covering all modules of the system. All tests are expected to pass if the system meets the specified requirements.

6.0 CONCLUSION

This test plan provides comprehensive coverage of all Smart Banklytics system modules. Each test case is designed to validate functional correctness, input handling, security enforcement, and data integrity. Successful execution of all 53 test cases confirms that the system meets the requirements specified in the SRS and is ready for deployment.